

Data Management Plan for BIOL530H Mini-Project

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Data Collection

This project involves a secondary analysis of a pre-existing dataset. No new raw data will be collected. The primary dataset is the Theoph pharmacokinetics dataset, openly available within the base R environment. The data is structured in a tabular format containing 132 observations of 5 different variables: weight (Wt), serum concentration (conc), Subject, dose, and time. Derived variables will be generated using R and saved in open, non-proprietary file formats to ensure long-term readability. Data will be organized into processed or raw data files, and limited analyses will be conducted in a single R script file. Files will be named following consistent formatting that is easily understood without an additional README file. Versioning will be conducted using GitHub and annotated commits will be uploaded for transparency.

Documentation and Metadata

The primary documentation for this project will be an annotated R script. In addition, a README file will be created that will define all variables, outline the file structure, and explain the methodology so the data can be correctly interpreted by others. Therefore, the data and analyses follow the FAIR data principles.

Storage, Backup, Data Security and Access Management

Data will be stored locally on the primary researcher's hard-drive and backed up on GitHub and OneDrive to prevent data loss. The total project is estimated to be less than 5GB, so additional storage is not required. This public dataset contains no personal identifiable information or sensitive health information, so it is classified as low risk. Therefore, strict access management, technical controls, or physical controls are not applicable.

Deposit, Reuse, and Preservation

Once completed, the full project including the analysis scripts, figures, README, and data dictionary files will be deposited into Open Science Framework (OSF) for discovery and reuse. Borealis will be used to archive and assign a DOI to the data. Data will be licensed using a CC-BY license as it is most open and flexible.

Responsibilities and Resources

The primary researcher is solely responsible for data management activities for the entire project, including writing the code, executing backups, and achieving the final project in Borealis. No additional personnel, funding, or computing resources are required since the dataset is small.

Ethics and Legal Compliance

The data used in this study is entirely open-source, previously de-identified, and publicly available. As it contains no sensitive information/data, methods of data anonymization are not required. There are no legal constraints, intellectual property issues, or FIPPA confidentiality risks. Therefore, an ethics board review is not required for this secondary analysis.